

Torque Ripple Suppression Method with Multilevel Inverter and Feedforward Dead-Time Compensation

Shuhei Shimmyo^{*a)} Member, Kazuaki Yuki^{*} Member
Kouhei Ohnishi^{**} Fellow

(Manuscript received Dec. 25, 2015, revised March 4, 2016)

This paper discusses the suppression of torque ripples in motor drives. Current ripples caused by pulse-width modulation with frequencies of 10–100 kHz can be reduced by using high-frequency switching. On the other hand, current ripples caused by dead-times with 6kth-order ripples of fundamental frequencies increase with the switching frequency. Therefore, there is a trade-off relationship between the two ripples. To compensate for dead-times, feedforward compensation is widely used. However, effective compensation cannot be achieved if the feedforward compensation values contain modeling errors. This paper therefore reveals that the robustness of the output torque against compensation errors can be improved by using multilevel inverters. The proposed method is validated through simulations and experimental results.

Keywords: equivalent carrier frequency, multilevel inverter, distortion index, feedforward, motor drive, dead-time compensation

1. Introduction

Brushless DC motors are widely used in servo systems, and precise motion control for robotics has been achieved. In recent times, many studies have focused on acceleration-based control^{(1)–(3)} and bilateral control^{(4)–(6)}. Bilateral control is a teleoperation method that transfers environmental haptic information to an operator. In addition, micro-macro bilateral control⁽⁷⁾ achieves the scaled motion of human skills (supported by experience and knowledge) by using scale gains; these controllers are constructed in oblique coordinates⁽⁸⁾. To scale humans' sophisticated motions, actuators that fully eliminate torque pulsation are required.

Traditionally, two-level inverters were widely used to drive brushless DC motors. In traditional two-level inverters, the DC voltage is divided by two switches to transform the phase voltage into a two-level waveform. Here, frequencies of 10–100 kHz are used to switch each power device by using pulse-width modulation (PWM). When using PWM, it is known that the torque ripple is strongly affected by the current ripple during low-load operation. The current becomes distorted primarily by the effects of voltage errors caused by dead-times⁽⁹⁾. The dead-times are widely used to prevent short-circuits between the positive and the negative sides, because power devices have turn-on and turn-off delay characteristics during switching. During dead-times, the output phase voltages are determined by the polarities of the output phase currents.

Generally, current ripples caused by PWM with frequencies of 10–100 kHz can be reduced by using high-frequency switching. On the other hand, current ripples caused by dead-times with 6kth-order ripples of fundamental frequencies increase with the switching frequencies. This is because dead-time voltage errors increase with the switching frequencies. Therefore, there is a trade-off relationship between the two ripples⁽¹⁰⁾.

Many dead-time compensation methods have been proposed for torque ripple suppression. Mainly, these can be labeled as “Feedforward Compensation” and “Feedback Compensation.” In feedforward dead-time compensation^{(11)–(13)}, it is necessary to obtain accurate polarity information of the current and accurate values of voltage errors. The widely used feedback dead-time compensation approach—for example, by using a disturbance observer (DOB)^{(14)–(16)}—does not require this information.

In recent times, studies have started focusing on multilevel inverters. Multilevel inverters can improve wave distortions as well as a motor's efficiency and control performance, including torque ripple. In our previous research, we evaluated feedback dead-time compensation by using DOB and the distortion indices of torque (DIT) for multilevel inverters⁽¹⁷⁾. Here, it was revealed that torque ripple could be suppressed by using multilevel inverters under the same feedback compensation; this means that the DOBs tested with two- and four-level inverters have the same observer gains. Although this study revealed that multilevel inverters were effective for torque ripple suppression, feedforward dead-time compensation was not evaluated.

Therefore, this study focused on feedforward dead-time compensation. In practice, the phase angle errors and modeling errors of voltage errors remain even if designers manage to tune the control parameters. Feedforward compensation is

a) Correspondence to: Shuhei Shimmyo. E-mail: shuhei.shimmyo@toshiba.co.jp

^{*} Power and Industrial Systems R&D Center, Toshiba Corporation 1, Toshiba-cho, Fuchu, Tokyo 183-8511, Japan

^{**} Department of System Design Engineering, Keio University 3-14-1, Hiyoshi, Kouhoku-ku, Yokohama 223-8522, Japan

therefore sensitive to modeling errors. Here, multilevel inverters are used for motor drives. By using these inverters, the robustness of the output torque against the phase angle errors and modeling errors of dead-times can be improved; furthermore, these inverters can improve the wave distortion, and therefore, the total torque ripples can be suppressed. The proposed method is validated through simulations and experimental results.

The remainder of this paper is organized as follows. In section 2, the model of the brushless DC motor is explained. In section 3, the voltage errors caused by dead-times are explained. In section 4, the DITs for multilevel motor drives are proposed. In section 5, the optimal carrier frequency is explained. In section 6, the simulations are discussed. In section 7, the experimental results are presented. Finally, the conclusions of this study are presented in section 8.

2. Model of Brushless DC Motor

In this section, a model of a brushless DC motor is presented. Generally, by applying the dq transformation represented by (1), the voltage equation can be simplified to (2). In this paper, the effective electric power is evaluated along the q axis.

$$C = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos \theta & \cos(\theta - \frac{2}{3}\pi) & \cos(\theta + \frac{2}{3}\pi) \\ \sin \theta & \sin(\theta - \frac{2}{3}\pi) & \sin(\theta + \frac{2}{3}\pi) \end{bmatrix} \cdots \cdots (1)$$

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} R_a + pL_d & -\omega L_q \\ \omega L_d & R_a + pL_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ \omega K_e \end{bmatrix} \cdots \cdots (2)$$

Here, R_a and L_o represent the resistance and self-inductance of each winding, respectively. Additionally, v_o , i_o , θ , and ω represent the voltage, current, electrical angle, and electrical angular velocity, respectively. $p = d/dt$ represents the differential operator and K_e , the speed constant.

3. Voltage Error Caused by Dead-Time

Here, the voltage errors caused by dead-times in general multi-level inverters are explained.

3.1 Voltage Error of Two-level Inverters

Figure 1(a) shows the circuit of the U phase of the two-level inverters. The ideal output voltage, modulation waveform, and voltage errors of the two-level inverters are shown in Fig. 1(b). Here, the phase delay of the output current is

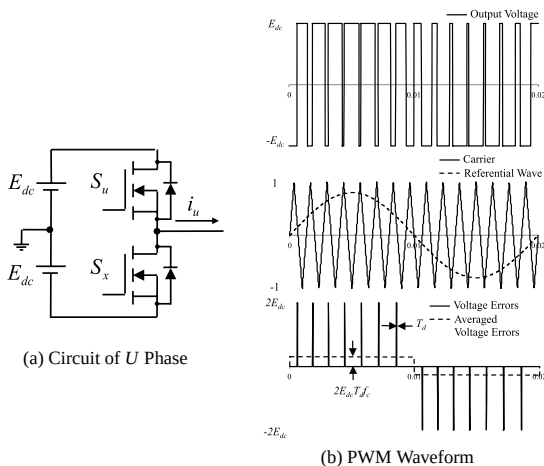


Fig. 1. Two-Level Inverters

assumed to be quite small. Furthermore, the power devices are treated as ideal switches, meaning that they do not have on- or off-delay characteristics. T_d denotes the dead-time and E_{dc} , the half-value of the DC voltage. When the output current has positive polarity during dead-time, the current flows to the diode of the lower switch. This means that the output voltage becomes $-E_{dc}$ during dead-time as long as the current does not become 0 A, and vice versa. By defining the carrier frequency f_c and switching period T_c , the averaged value of the voltage errors can be represented as follows:

$$v_u^{err}(i_u) = \frac{1}{T_c} (v_u^{ideal} - v_u^{prac}(i_u)) T_d \\ = 2E_{dc} T_d f_c \text{sgn}(i_u). \cdots \cdots (3)$$

3.2 Voltage Error of Multilevel Inverters

To explain our method more simply, an example of a four-level inverter is described here. Figure 2(a) shows the circuit of the U phase of the four-level inverters. The ideal output voltage, modulation waveform, and voltage errors of the four-level inverters are shown in Fig. 2(b). Here, the phase delay of the output current is also assumed to be quite small, and PWM is performed using the carrier-shifted method. v_u^{max} defines the maximum values of the output voltage during dead-times, and v_u^{min} defines the minimum values of the output voltage during dead-times. When the output current has positive polarity during dead-time, the output voltage becomes v_u^{min} during dead-time as long as the current does not become 0 A, and vice versa. Here, v_u^{max} and v_u^{min} in the four-level inverters are given as follows:

$$v_u^{max} = \begin{cases} E_{dc} & (\frac{1}{3} \leq v_u^{ref} \leq 1) \\ \frac{E_{dc}}{3} & (-\frac{1}{3} \leq v_u^{ref} < \frac{1}{3}) \\ -\frac{E_{dc}}{3} & (-1 \leq v_u^{ref} < -\frac{1}{3}) \end{cases} \cdots \cdots (4)$$

$$v_u^{min} = \begin{cases} \frac{E_{dc}}{3} & (\frac{1}{3} \leq v_u^{ref} \leq 1) \\ -\frac{E_{dc}}{3} & (-\frac{1}{3} \leq v_u^{ref} < \frac{1}{3}) \\ -E_{dc} & (-1 \leq v_u^{ref} < -\frac{1}{3}). \end{cases} \cdots \cdots (5)$$

Here, v_u^{ref} denotes the reference modulation wave of the PWM. v_u^{max} and v_u^{min} in the multilevel inverters are given by (6) and (7), respectively. Here, m denotes the number of levels.

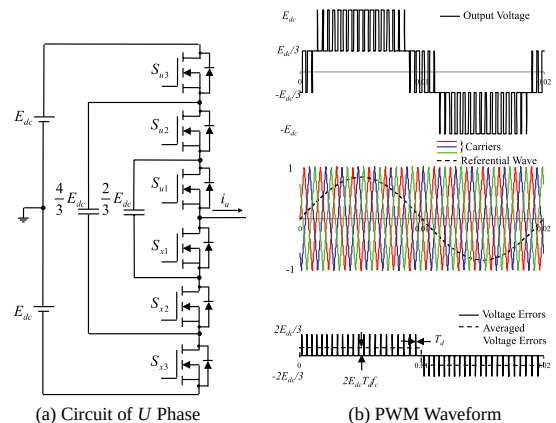


Fig. 2. Four-Level Inverters

$$v_u^{max} = \begin{cases} E_{dc} & (\frac{m-3}{m-1} \leq v_u^{ref} \leq 1) \\ \frac{m-3}{m-1} E_{dc} & (\frac{m-5}{m-1} \leq v_u^{ref} < \frac{m-3}{m-1}) \\ \vdots & \\ -\frac{m-3}{m-1} E_{dc} & (-1 \leq v_u^{ref} < -\frac{m-3}{m-1}) \end{cases} \dots\dots (6)$$

$$v_u^{min} = \begin{cases} \frac{m-3}{m-1} E_{dc} & (\frac{m-3}{m-1} \leq v_u^{ref} \leq 1) \\ \frac{m-5}{m-1} E_{dc} & (\frac{m-5}{m-1} \leq v_u^{ref} < \frac{m-3}{m-1}) \\ \vdots & \\ -E_{dc} & (-1 \leq v_u^{ref} < -\frac{m-3}{m-1}) \end{cases} \dots\dots\dots (7)$$

Additionally, the practical voltage of multilevel inverters during dead-time can be represented as (8).

$$v_u^{prac}(i_u) = \begin{cases} v_u^{min} & (i_u > 0) \\ v_u^{max} & (i_u < 0) \end{cases} \dots\dots\dots (8)$$

In the case of multilevel inverters that are modulated using the carrier-shifted method, dead-times are generated $m - 1$ times more frequently than in two-level inverters within the switching period T_c . The averaged value of the voltage errors for generalized multilevel inverters can be represented as follows:

$$\begin{aligned} v_u^{err}(i_u) &= \frac{1}{T_c} (v_u^{ideal} - v_u^{prac}(i_u))(m-1)T_d \\ &= \frac{1}{T_c} |v_u^{max} - v_u^{min}|(m-1)T_d \text{sgn}(i_u) \\ &= \frac{2E_{dc}}{m-1} (m-1)T_d f_c \text{sgn}(i_u) \\ &= 2E_{dc}T_d f_c \text{sgn}(i_u). \dots\dots\dots (9) \end{aligned}$$

Therefore, the averaged voltage error does not depend on the number of levels. On the other hand, the equivalent carrier frequency of the multilevel inverters can be represented as follows:

$$f^{eq} = (m-1)f_c. \dots\dots\dots (10)$$

From the viewpoint of the motor's control performance and efficiency, it is better to use high equivalent carrier frequencies f^{eq} . By substituting (10) into (9), the averaged value of the voltage error can be rewritten as follows:

$$v_u^{err}(i_u) = 2E_{dc}T_d \frac{f^{eq}}{m-1} \text{sgn}(i_u). \dots\dots\dots (11)$$

Equation (11) shows that the multilevel inverters can reduce the averaged voltage errors. In other words, a lower carrier frequency f_c for each switch can be obtained by increasing the number of levels m .

4. Distortion Index of Torque for Multilevel Motor Drives

In this section, the DIT for multilevel motor drives is proposed. First, the theoretical harmonics of multilevel inverters along the dq axis are derived. Second, the theoretical harmonics of the averaged dead-time voltage errors are derived. Third, the DIT taking the sensitivity function into account is presented. Fourth, the DIT taking the feedforward compensation into account is proposed.

4.1 Theoretical Harmonics of Multilevel Inverters along dq Axis The theoretical harmonics of m -level inverters driven using the carrier-shifted method are represented as follows ⁽¹⁸⁾⁽¹⁹⁾:

$$\begin{aligned} v_{out}(\varphi) &= aE_{dc} \sin(\omega_0 t + \varphi) + \sum_{n=1}^{\infty} \frac{4E_{dc}}{n(m-1)\pi} \left[\right. \\ &2 \sum_{l=1}^{\infty} J_{2l-1}(a\lambda) \sin\{(2l-1)(\omega_0 t + \varphi)\} \cos(\lambda) \\ &\left. - \{J_0(a\lambda) + 2 \sum_{l=1}^{\infty} J_{2l}(a\lambda) \cos\{2l(\omega_0 t + \varphi)\} \sin(\lambda)\} \right] \\ &\cdot \cos(n(m-1)\omega_c t). \dots\dots\dots (12) \end{aligned}$$

Here, $\lambda = n(m-1)\pi/2$ is defined. ω_0 and ω_c represent the angular velocity of the fundamental wave and of the carrier wave, respectively. Additionally, J_k refers to the k th-order Bessel function of the first kind, and a indicates the modulation factor. From these relationships, the spectra of the $n(m-1)f_c \pm kf_0$ components are represented as follows:

$$V_{n(m-1)f_c \pm kf_0} = \frac{4E_{dc}}{n(m-1)\pi} J_k(a\lambda). \dots\dots\dots (13)$$

Here, $n(m-1) = 1, 3, 5, \dots$ when $k = 0, 2, 4, \dots$, and $n(m-1) = 2, 4, 6, \dots$ when $k = 1, 3, 5, \dots$. By applying the dq transform represented as (1), the theoretical harmonics of the multi-level inverters along the dq axis can be obtained. The theoretical harmonics of multilevel inverters along the d axis are represented as follows:

$$\begin{aligned} v_d &= \sum_{n=1}^{\infty} \frac{4E_{dc}}{n(m-1)\pi} \sqrt{\frac{3}{2}} \left[\right. \\ &-2 \sum_{l=1}^{\infty} \{J_{6l-4}(a\lambda) + J_{6l-2}(a\lambda)\} \sin(\lambda) \cos((6l-3)\omega_0 t) \\ &+ 2 \sum_{l=1}^{\infty} \{J_{6l+1}(a\lambda) + J_{6l-1}(a\lambda)\} \cos(\lambda) \sin(6l\omega_0 t) \left. \right] \\ &\cdot \cos(n(m-1)\omega_c t). \dots\dots\dots (14) \end{aligned}$$

As with the d axis, the theoretical harmonics along the q axis can be represented as follows:

$$\begin{aligned} v_q &= \sqrt{\frac{3}{2}} aE_{dc} + \sum_{n=1}^{\infty} \frac{4E_{dc}}{n(m-1)\pi} \sqrt{\frac{3}{2}} \left[\right. \\ &-2 \sum_{l=1}^{\infty} \{J_{6l-4}(a\lambda) - J_{6l-2}(a\lambda)\} \sin(\lambda) \sin((6l-3)\omega_0 t) \\ &+ J_1(a\lambda) + 2 \sum_{l=1}^{\infty} \{J_{6l+1}(a\lambda) - J_{6l-1}(a\lambda)\} \cos(\lambda) \\ &\left. \cdot \cos(6l\omega_0 t) \right] \cos(n(m-1)\omega_c t). \dots\dots\dots (15) \end{aligned}$$

From these relationships, the spectra of the $n(m-1)f_c \pm kf_0$ components along the dq axis can be represented as follows:

$$V_{d(n(m-1)f_c \pm kf_0)} = \frac{2\sqrt{6}E_{dc}}{n(m-1)\pi} \{J_{k+1}(a\lambda) + J_{k-1}(a\lambda)\} \dots\dots\dots (16)$$

$$V_{q(n(m-1)f_c \pm kf_0)} = \frac{2\sqrt{6}E_{dc}}{n(m-1)\pi} \{J_{k+1}(a\lambda) - J_{k-1}(a\lambda)\} \dots \dots \dots (17)$$

Here, $n(m-1) = 1, 3, 5, \dots$ when $k = 6l - 3$, and $n(m-1) = 2, 4, 6, \dots$ when $k = 6l$. In particular, for $k = 0$, V_q is represented as follows:

$$V_{q(n(m-1)f_c)} = \frac{2\sqrt{6}E_{dc}}{n(m-1)\pi} J_1(a\lambda) \dots \dots \dots (18)$$

As shown in (16) and (17), the theoretical harmonics along the d axis are expressed as the common mode of J_{k+1} and J_{k-1} . The theoretical harmonics along the q axis are expressed as the differential mode of J_{k+1} and J_{k-1} .

4.2 Theoretical Harmonics of Averaged Dead-Time Voltage Error Here, the theoretical harmonics of the averaged dead-time voltage error are derived. As mentioned above, dead-time voltage errors can be represented as (11). Therefore, by applying the Fourier series expansion of the rectangular waveform, the averaged dead-time errors in the U phase can be represented as follows:

$$v_u^{err} = 2E_{dc}T_d \frac{f^{eq}}{m-1} \sum_{k=1}^{\infty} \frac{4}{(2k-1)\pi} \sin((2k-1)\omega_0 t) \dots \dots \dots (19)$$

The theoretical harmonics of the averaged dead-time voltage error of multilevel inverters along the dq axis are derived using (1) as follows:

$$v_d^{err} = \frac{4\sqrt{6}E_{dc}T_d f^{eq}}{(m-1)\pi} \left[\sum_{k=1}^{\infty} (\Gamma_k^+ + \Gamma_k^-) \sin(6k\omega_0 t) \right] \dots \dots \dots (20)$$

$$v_q^{err} = \frac{4\sqrt{6}E_{dc}T_d f^{eq}}{(m-1)\pi} \left[1 + \sum_{k=1}^{\infty} (\Gamma_k^+ - \Gamma_k^-) \cos(6k\omega_0 t) \right] \dots \dots \dots (21)$$

Here, $\Gamma_k^- = \frac{1}{6k-1}$ and $\Gamma_k^+ = \frac{1}{6k+1}$ are defined. From these relationships, the spectra of the $6kf_0$ components along the dq axis are represented as follows:

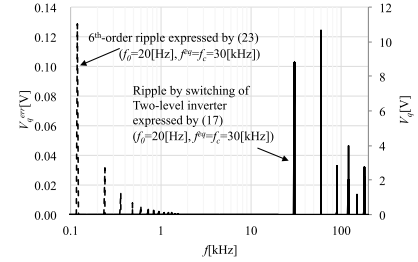
$$V_{d(6kf_0)}^{err} = \frac{4\sqrt{6}E_{dc}T_d f^{eq}}{(m-1)\pi} (\Gamma_k^+ + \Gamma_k^-) \dots \dots \dots (22)$$

$$V_{q(6kf_0)}^{err} = \frac{4\sqrt{6}E_{dc}T_d f^{eq}}{(m-1)\pi} (\Gamma_k^+ - \Gamma_k^-) \dots \dots \dots (23)$$

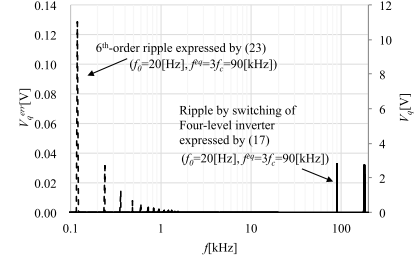
In particular, for $k = 0$, $V_{q(0)}^{err}$ is represented as follows:

$$V_{q(0)}^{err} = \frac{4\sqrt{6}E_{dc}T_d f^{eq}}{(m-1)\pi} \dots \dots \dots (24)$$

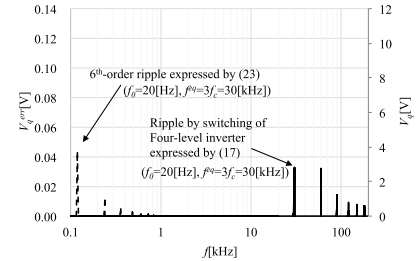
In the theoretical harmonics of the averaged dead-time voltage errors along the dq axis, the common mode of Γ_k^- and Γ_k^+ appeared along the d axis, and the differential mode of Γ_k^- and Γ_k^+ appeared along the q axis. Figure 3 shows the schematics of the theoretical harmonics of the multilevel inverters and the theoretical harmonics of the averaged dead-time voltage errors. From Fig. 3(a) and Fig. 3(b), it is clear that the spectra of the switching can be reduced by using four-level inverters.



(a) Spectra of Two-level Inverter ($f^{eq} = f^c = 30$ kHz)



(b) Spectra of Four-level Inverter ($f^{eq} = 3f^c = 90$ kHz)



(c) Spectra of Four-level Inverter ($f^{eq} = 3f^c = 30$ kHz)

Fig. 3. Schematics of Theoretical Harmonics of Voltage along q axis (without any compensation, $f_0 = 20$ Hz, $2E_{dc} = 48$ V, $T_d = 1 \mu s$, $a = 1$)

In addition, from Fig. 3(c), the spectra of the averaged dead-time errors can be reduced by lowering the carrier frequency f_c . Here, the equivalent carrier frequency f^{eq} in Fig. 3(c) is same as that in Fig. 3(a); this means that lowering the carrier frequency does not increase the current ripples caused by PWM.

4.3 Distortion Index of Torque Taking Sensitivity Function Into Account

This study focuses on a brushless DC motor. This motor is a so-called surface permanent magnet synchronous motor (SPMSM), which have no saliencies. The output torque of the SPMSM can be represented as follows:

$$T = K_t i_q \dots \dots \dots (25)$$

Here, K_t denotes the torque coefficient. Therefore, the torque ripple of the $n(m-1)f_c \pm kf_0 = n f^{eq} \pm kf_0$ components caused by the PWM waveform are represented as follows:

$$\Delta T_{(n f^{eq} \pm kf_0)} = \frac{K_t V_{q(n f^{eq} \pm kf_0)}}{\sqrt{(2\pi(n f^{eq} \pm kf_0))^2 L_q^2 + R_a^2}} \dots \dots \dots (26)$$

Additionally, the torque ripple of the $6kf_0$ components caused by the dead-times is represented as follows:

$$\Delta T_{(6kf_0)} = \frac{K_t |G_S(j2\pi 6kf_0)| V_{q(6kf_0)}^{err}}{\sqrt{(2\pi 6kf_0)^2 L_q^2 + R_a^2}} \dots \dots \dots (27)$$

Here, the sensitivity function can be represented as follows:

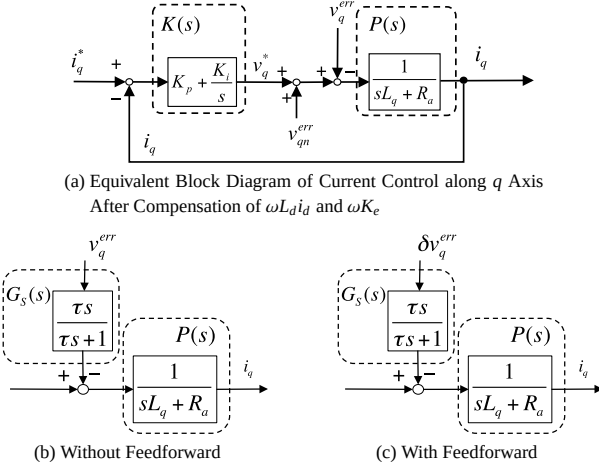


Fig. 4. Schematics of Sensitivity Functions of PI Controller

$$G_S(s) = \frac{1}{1 + P(s)K(s)} \quad (28)$$

Here, s denotes the Laplace operator. $P(s)$ and $K(s)$ denote the control plant and arbitrary controller, respectively. Finally, the DIT taking the sensitivity function into account can be defined as follows:

$$\text{DIT} = \frac{1}{T_0} \sqrt{\sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \left(\frac{\Delta T_{(n f_{eq} \pm k f_0)}}{\sqrt{2}} \right)^2 + \sum_{k=0}^{\infty} \left(\frac{\Delta T_{(6k f_0)}}{\sqrt{2}} \right)^2} \quad (29)$$

Here, T_0 denotes the fundamental torque.

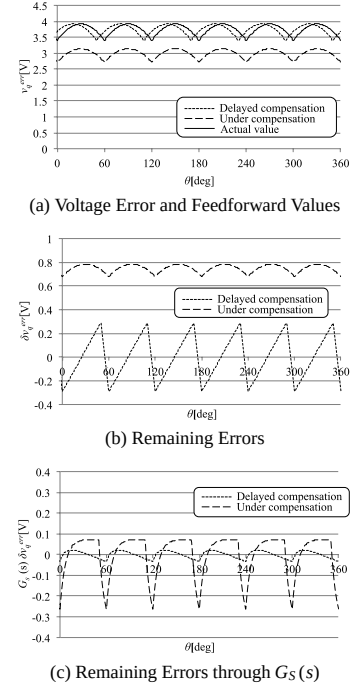
4.4 Distortion Index of Torque Taking Feedforward Compensation into Account Here, a DIT that takes the feedforward compensation into account is discussed. To compensate for dead-time voltage errors, feedforward compensation is widely used^{(11)–(13)}. Sukegawa et al. proposed feedforward compensation in Park coordinates⁽¹¹⁾. In this paper, to explain the torque ripple clearly, feedforward compensation in Park coordinates is discussed. If we choose suitably high PI gains, the output waveforms will be improved. However, it is difficult to improve the output waveforms over the entire operational range (several speed conditions) by using only a PI controller, because the sensitivity function of the PI controller is constant although the frequency of the $6k$ -th order components varies depending on the rotational speed of the motor. Therefore, feedforward compensation is effective. The voltage equation along the q axis, including the voltage errors, can be represented as follows:

$$v_q = L_q \frac{di_q}{dt} + R_a i_q + \omega L_d i_d + \omega K_e + v_q^{err} \quad (30)$$

Additionally, the voltage errors caused by dead-times can be calculated by (31). Actually, the theoretical voltage errors caused by dead-times are calculated along the dq axis; the infinite order harmonics can be calculated as follows:

$$\begin{bmatrix} v_{dn}^{err} \\ v_{qn}^{err} \end{bmatrix} = C \begin{bmatrix} v_{un}^{err} \\ v_{vn}^{err} \\ v_{wn}^{err} \end{bmatrix} \quad (31)$$

Here, the index o_n denotes the nominal value. By substituting the nominal DC voltage $2E_{dcn}$, nominal dead-time T_{dn} ,

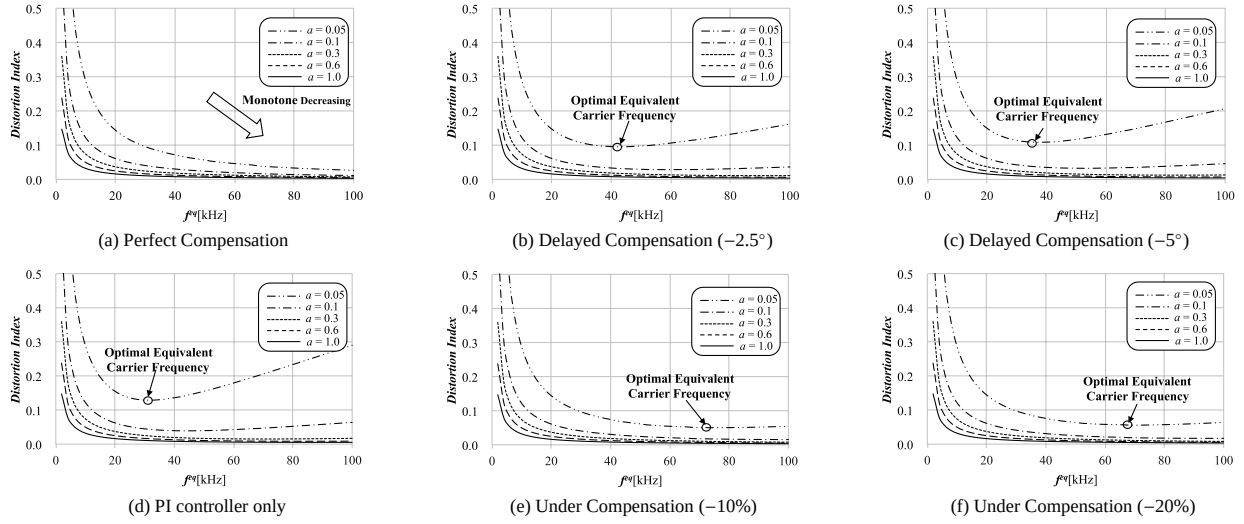
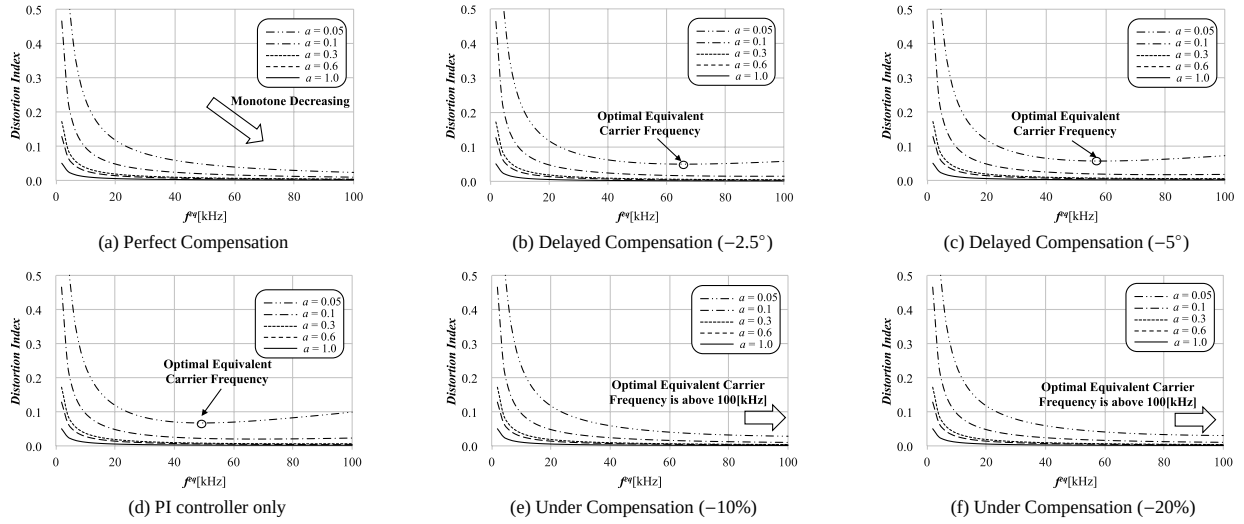

 Fig. 5. Schematics of Dead-time Compensation with Feedforward along q axis

nominal carrier frequency f_{cn} , and nominal polarity of the phase current $\text{sgn}(i_{on})$ in (3), v_{un}^{err} , v_{vn}^{err} , and v_{wn}^{err} are obtained. Figure 5 shows the theoretical voltage errors caused by dead-time along the dq axis, including the infinite order harmonics. By substituting (31) into (30) and applying feedforward compensation for interference terms, the voltage equation for the q axis can be represented as (32), if the calculated interference terms $\omega L_d i_d$ and ωK_e are perfectly matched to the actual values.

$$\begin{aligned} v_q &= L_q \frac{di_q}{dt} + R_a i_q + v_q^{err} - v_{qn}^{err} \\ &= L_q \frac{di_q}{dt} + R_a i_q + \delta v_q^{err} \quad (32) \end{aligned}$$

Here, δv_q^{err} denotes the remaining errors caused by dead-times along the q axis after feedforward compensation. Ideally, $\delta v_q^{err} = 0$ is achieved if there are no modeling errors. For small voltage compensation, the voltage drops of the power devices may be indispensable. These voltage drops are caused by the conduction resistance $R_{DS(on)}$ of the switching devices during conduction or by the forward voltage of the diodes V_f during freewheeling. These are the main reasons to generate the modeling errors δv_q^{err} .

Figure 4 shows the schematics of the sensitivity function by using the PI controller. Figure 4(a) shows the equivalent block diagram of the current control along the q axis after the compensation of $\omega L_d i_d$ and ωK_e . In this study, the sensitivity function is treated as a first-order system by choosing $K_p = \frac{L_q}{\tau}$ and $K_i = \frac{R_a}{\tau}$ to simplify the analysis. By using the above-described procedures, we can obtain the output current along the q axis as shown in Figs. 4(b) and 4(c), which show the case without and with feedforward compensation, respectively. Although only the schematics of the sensitivity functions along the q axis are shown here, the same procedures can be applied along the d axis. Figure 5 shows the voltage


 Fig. 6. Distortion Indices of Torque of Two-level Inverters ($f_0 = 20$ Hz, $2E_{dc} = 48$ V, $T_d = 1 \mu\text{s}$, $\tau = 0.5$ ms)

 Fig. 7. Distortion Indices of Torque of Four-level Inverters ($f_0 = 20$ Hz, $2E_{dc} = 48$ V, $T_d = 1 \mu\text{s}$, $\tau = 0.5$ ms)

errors and feedforward values along the q axis. As mentioned above, because phase angle errors and modeling errors of the voltage errors exist, it is impossible to compensate them perfectly. In this figure, the solid line denotes the actual voltage errors caused by the dead-times. The dashed line denotes the feedforward values under compensation (-20%). The dotted line denotes the feedforward values under delayed compensation (-10°). Here, although worst-case scenarios are shown, it is clear that the phase errors and modeling errors generate remaining errors δv_q^{err} , and these distort the waveform. The $6kf_0$ components of the torque ripple are represented as follows:

$$\delta \Delta T_{(6kf_0)} = \frac{K_t |G_S(j2\pi 6kf_0)| \delta v_{q(6kf_0)}^{err}}{\sqrt{(2\pi 6kf_0)^2 L_q^2 + R_a^2}} \dots \dots \dots (33)$$

Finally, the DIT that takes the feedforward compensation into account is represented as follows:

$$\text{DIT} = \frac{1}{T_0} \sqrt{\sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \left(\frac{\Delta T_{(n f_{eq} \pm k f_0)}}{\sqrt{2}} \right)^2 + \sum_{k=0}^{\infty} \left(\frac{\delta \Delta T_{(6kf_0)}}{\sqrt{2}} \right)^2} \dots \dots \dots (34)$$

5. Optimal Equivalent Carrier Frequency

In this section, the numerical analysis of the optimal equivalent carrier frequency is presented. This study used an EC-4pole brushless DC motor (305015) from Maxon Corp. This motor did not have any uneven magnetic distributions because of its structure. Therefore, a sinusoidal wave drive is suitable for this motor. The hardware and control parameters are shown in Table 1. Figures 6 and 7 show the DITs of the two- and four-level inverters, respectively. The DITs were calculated using (29) and (34). Here, the fundamental torque is calculated using $T_0 = K_t / R_a (\sqrt{3}/2 a E_{dc} - \omega_m P_n K_e)$. Thus, it is assumed that the fundamental torque is increased by increasing the modulation factor a , because the rotational speed ω_m is assumed to be a constant value. Ideally, infinite harmonics should be considered; these are shown in Figs. 6 and 7. However, only finite harmonics can be considered in a practical calculation. From the theoretical harmonics of the q axis (17), it is clear that the amplitudes of the high-order harmonics are lowered. Therefore, above 400 kHz, we assume that the current ripple is very small (indeed, negligible) in this study. Therefore, only frequencies below 400 kHz were

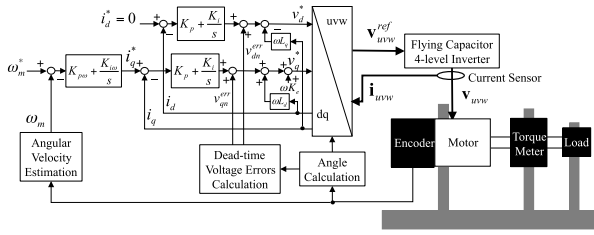


Fig. 8. Schematics of Entire Control System

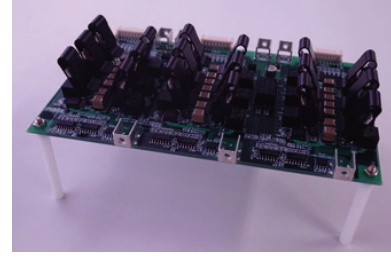


Fig. 9. Flying Capacitor Four-level Inverter

considered.

Figures 6 and 7 clearly show that the DITs could be reduced by using multilevel inverters. Additionally, the optimal equivalent carrier frequency was changed based on the modulation factor a . The DITs increased when the modulation factor was small; this means that the torque ripple ratio increases in the low-load region when the modulation factor is low. From these figures, it is clear that the DITs of the optimal equivalent carrier frequency can be reduced by using a multilevel inverter. This means that the multilevel inverters make the output torque robust against the compensation errors caused by the modeling errors of the voltage errors and phase errors.

6. Simulations

This section presents the simulation results. Figure 8 shows the entire control system. Table 1 shows the hardware and control parameters.

Figure 10 shows the simulation results of only the four-level inverters. In these simulations, the equivalent carrier frequency was selected from Fig. 7 as 75 kHz, because compensation delay within -2.5° and compensation amplitude error within -10% may be achieved for practical implementation. These results show that the torque ripple could be suppressed by using feedforward compensation. The torque ripple clearly increased with delayed compensation and under compensation.

As mentioned above, the 6 k -th order harmonics are not only affected by the voltage errors caused by the dead-times but also by the sensitivity function of the controller. In particular, 6th-order components with comparatively high amplitude were observed in the simulation results shown in Fig. 10(c). In the theory, only the sensitivity function of the current control loop was considered, and the variation of the capacitor voltage was not considered: they were treated as perfect voltage sources. On the other hand, the outer loop was modeled, and the variation of the capacitor voltage was also considered in the simulation. These seemed to be the main reasons for the difference between the theory and the simulations. Except for the 6th-order components in Fig. 10(c), the value of each spectrum of the theory and the simulation results was the same. The theory was validated from these simulation results.

7. Experimental Results

This section presents the experimental results. Figure 9 shows the tested flying-capacitor four-level inverter. The control system and all parameters were the same as in the simulations.

The experimental results of the two-level inverters are

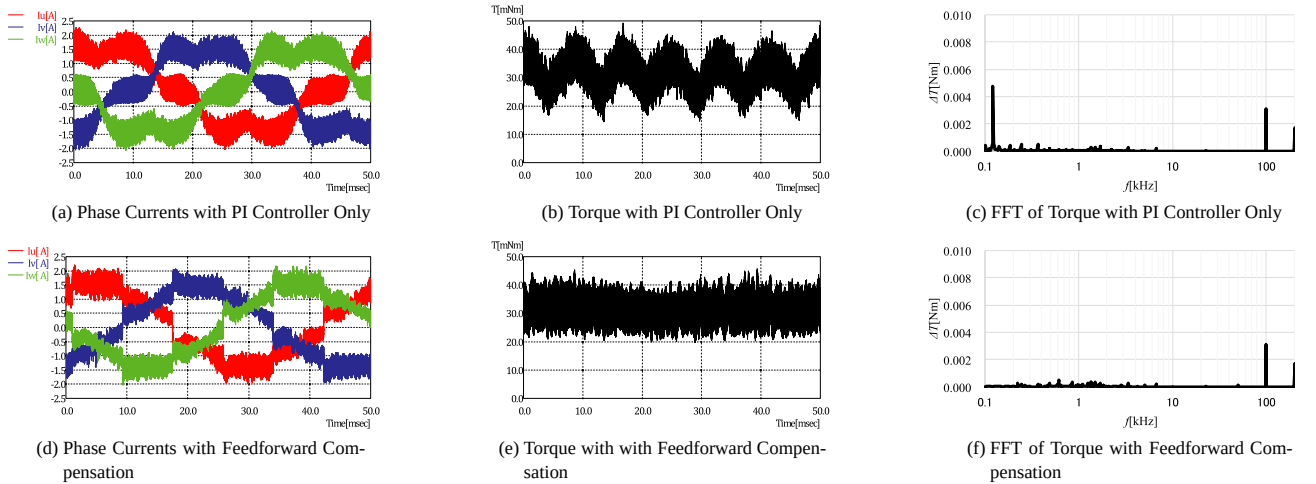
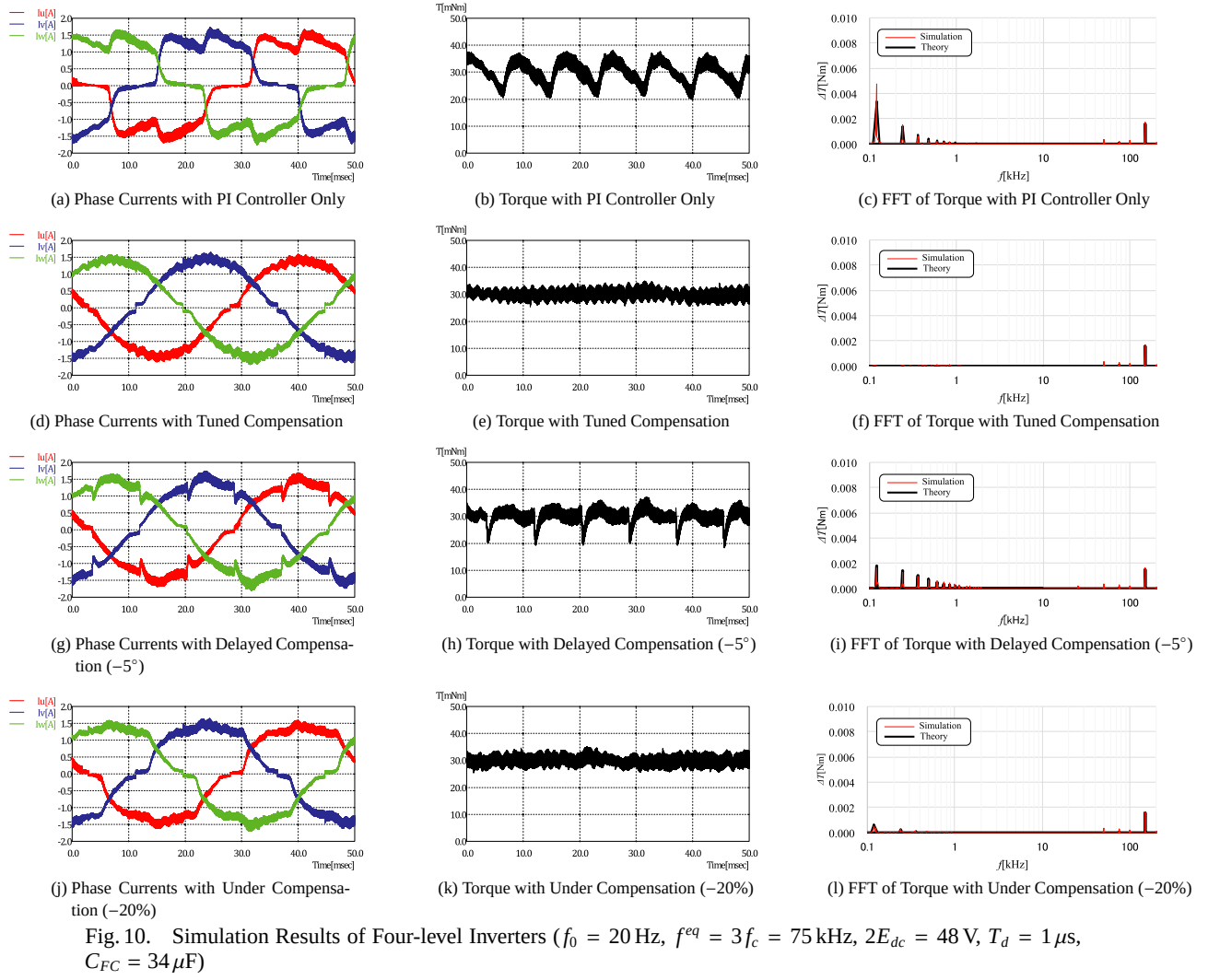
Table 1. Hardware and Control Parameters

Proportional gain of current control [V/A]	K_p	0.065
Integration gain of current control [V/A]	K_i	386.0
Proportional gain of velocity control [As/rad]	$K_{p\omega}$	3.13
Integration gain of velocity control [As/rad]	$K_{i\omega}$	5.49
LPF cut-off frequency for each sensor [Hz]	f_{LPF}	2000
DC voltage [V]	$2E_{dc}$	48.0
Velocity command [rad/s]	ω_m^*	62.8
Load torque [mNm]	$T^{const.}$	30
Capacitance of flying-capacitor [μ F]	C_{FC}	34.0
Torque constant [mNm/A]	K_t	17.7
Armature resistance [Ω]	R_a	0.193
Motor inductance [μ H]	L_d, L_q	32.5
Speed constant [mVs/rad]	K_e	8.45
Pole pair number	P_n	2

shown in Fig. 11. In these experiments, the equivalent carrier frequency was selected from Fig. 6 as 50 kHz, because compensation delay within -2.5° and compensation amplitude error within -10% may be achieved for practical implementation. The experimental results of the four-level inverters are shown in Fig. 12. In these experiments, the equivalent carrier frequency was selected as 75 kHz for the same reasons as above.

Here, the output torque was calculated by using the torque constant and current along the q axis by (25), because the torque meter has a narrow bandwidth to measure the torque. i_q was calculated by using the phase currents measured by the oscilloscope. Here, the sampling frequency of the oscilloscope was set as 20 MHz to measure the high-order harmonics. These results show that the torque ripple can be suppressed by using feedforward compensation. A comparison of Figs. 10(c) and 12(c) shows that the high-order harmonics were slightly larger than those in the theory and simulation results. This may be caused by the modeling errors of the motor's constant and some voltage drops caused by the parasitic resistance of the wiring.

In addition, 6th-order components with comparatively low amplitude relative to the theory were observed in the experimental results shown in Fig. 12(c). These seemed to be affected by the modeling errors, as in the simulations. Furthermore, the actual motor impedance can be different from the nominal ones used in the experiments. If the motor impedances vary, the sensitivity function also varies. These seemed to be the main reasons for the difference between the theory and the experiments. In this research, Si MOSFETs with low conduction resistance $R_{DS(on)} = 10.7$ [m Ω] because of the low breakdown voltage of 200 V were used. This device's switching delay time of 10–100 ns is comparatively shorter than the dead time (1 μ s). Therefore, we can effectively evaluate the theory from the experimental results.



Although we can reduce the dead-time, the dead-time was set as $1 \mu s$ to confirm the effects of dead-time compensation and multilevel inverters.

In addition, the DITs of the theory and experimental results are calculated, and they are shown in Fig. 13. These figures clearly show that the DITs of the theory and experimental results have the same tendency. Although they are not perfectly

matched, the experimented points could be plotted near the theoretical curves. Furthermore, the DITs of the experimental results can be reduced by using feedforward compensation. In particular, the suppression effect by feedforward compensation in the two-level inverter is more remarkable than that in the four-level inverter. In other words, two-level inverters with feedforward compensation may show large

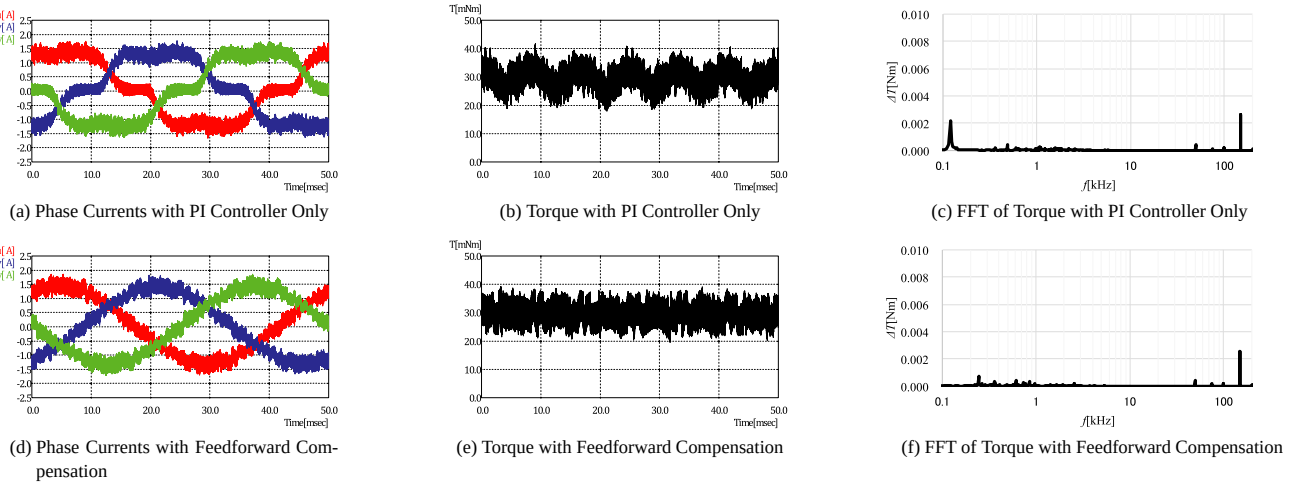


Fig. 12. Experimental Results of Four-level Inverters ($f_0 = 20$ Hz, $f^{eq} = 3f_c = 75$ kHz, $2E_{dc} = 48$ V, $T_d = 1$ μ s)

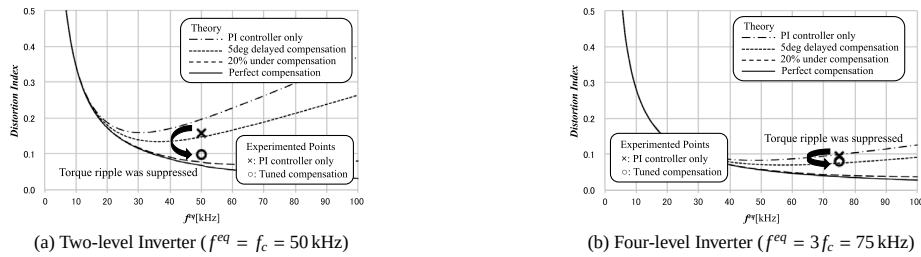


Fig. 13. Distortion Indices of Torque of Theory and Experimental Results ($a = 0.047$)

modeling errors if the phase angle errors and modeling errors of the dead-times remain.

From the above discussions, although there were some differences between the theory and the experimental results, these differences were mainly generated from the modeling errors. The tendency of the theory and experimental results was the same. The proposed method (combination of multilevel inverters and feedforward compensation) was therefore validated through the experimental results.

8. Conclusions

In this paper, we focused on feedforward dead-time compensation and the optimal carrier frequencies of multilevel motor drives. In practice, the phase angle errors and modeling errors of the dead-times remained, even if the designers managed to tune the control parameters. Here, multilevel inverters were used for the motor drives. By using these inverters, the robustness of the output torque against phase angle errors and modeling errors could be improved. Furthermore, these inverters could improve the wave distortion, and therefore, the total torque ripples could be suppressed. The proposed method was validated through simulations and experimental results.

References

- (1) K. Ohnishi, M. Shibata, and T. Murakami: "Motion Control for Advanced Mechatronics", *IEEE/ASME Trans. Mechatronics*, Vol.1, No.1, pp.56–67 (1996)
- (2) T. Murakami, F. Yu, and K. Ohnishi: "Torque Sensor Less Control in Multidegree-Of-Freedom Manipulator", *IEEE Trans. Ind. Electron.*, Vol.40, No.2, pp.259–265 (1993)
- (3) S. Katsura, Y. Matsumoto, and K. Ohnishi: "Analysis and Experimental Validation of Force Bandwidth for Force Control", *IEEE Trans. Ind. Electron.*, Vol.53, No.3, pp.922–928 (2006)
- (4) S. Katsura, Y. Matsumoto, and K. Ohnishi: "Realization of "Law of Action and Reaction" by Multilateral Control", *IEEE Trans. Ind. Electron.*, Vol.52, No.5, pp.1196–1205 (2005)
- (5) W. Iida and K. Ohnishi: "Reproducibility and Operationality in Bilateral Teleoperation", *IEEE Int. Work. Adv. Motion Control*, pp.212–222 (2004)
- (6) H. Tanaka, K. Ohnishi, H. Nishi, T. Kawai, Y. Morikawa, S. Ozawa, and T. Furukawa: "Implementation of Bilateral Control System Based on Acceleration Control Using FPGA for Multi-DOF Haptic Endoscopic Surgery Robot", *IEEE Trans. Ind. Electron.*, Vol.56, No.3, pp.618–627 (2005)
- (7) S. Susa, K. Natori, and K. Ohnishi: "Three-Chanel Micro-Macro Bilateral Control System with Scaling of Control Gains", in *Proc. IEEE 34th Annu. Conf. Ind. Electron. Soc.*, pp.2598–2603 (2008)
- (8) S. Sakaino, S. Sato, and K. Ohnishi: "Multi-DOF Micro-Macro Bilateral Controller Using Oblique Coordinate Control", *IEEE Trans. Ind. Inform.*, Vol.7, No.3, pp.446–454 (2011)
- (9) K. Yamamoto and K. Shinohara: "Analysis of AC Servo Motor Driven by PWM Inverter with Switching Dead-Time and Compensation for Output Voltage Deviation", *IEEJ Trans. IA*, Vol.116, No.9, pp.924–933 (1996) (in Japanese)
- (10) Y. Murai and T. Watanabe: "Waveform Distortion and Correction Circuit for PWM Inverters with Switching Lag-Times", *IEEE Trans. on Ind. Appl.*, Vol.1A-23, No.5, pp.881–885 (1987)
- (11) T. Sukegawa, K. Kamiyama, K. Mizuno, T. Matsui, and T. Okuyama: "Fully Digital Vector-Controlled PWM VSI Fed AC Drives with an Inverter Dead-Time Compensation Strategy", *IEEE Trans. on Ind. Appl.* Vol.27, No.3, pp.552–559 (1991)
- (12) J.W. Choi and S.K. Sul: "Inverter Output Voltage Synthesis Using Novel Dead Time Compensation", *IEEE Trans. on Power Electronics*, Vol.11, No.2, pp.221–227 (1996)
- (13) A.R. Munoz and T.A. Lipo: "On-line Dead-Time Compensation Technique for Open-Loop PWM-VSI Drive", *IEEE Trans. on Power Electronics*, Vol.14, No.4, pp.683–689 (1999)
- (14) H.S. Kim, H.T. Moon, and M.J. Youn: "On-line Dead-Time Compensation Method Using Disturbance Observer", *IEEE Trans. on Power Electronics*, Vol.18, No.6, pp.1136–1345 (2003)
- (15) N. Urasaki, T. Senjyu, K. Uezato, and T. Funabashi: "An Adaptive Dead-Time Compensation Strategy for Voltage Source Inverter Fed Motor Drives", *IEEE Trans. on Power Electronics*, Vol.20, No.5 (2005)
- (16) J. Itoh and T. Hoshino: "Analysis of Dead-time Error Correction Properties

for V/f Control with Disturbance Observer”, Proc. of IEEJapan IAS 2006, Vol.1, pp.179–182 (2006) (in Japanese)

- (17) S. Shimmyo, K. Takeuchi, N. Takahashi, M. Matsushita, and K. Ohnishi: “Multi-level Motor Drives for Torque Ripple Suppression Taking Control Sensitivity into Account”, *IEEE Trans. IA* Vol.5, No.2, pp.69–77 (2016)
- (18) I. Takahashi and S. Miyairi: “Relation between the Output Voltage: Waveform of the PWM Inverter and its Gate Control Signals”, *IEEE Trans. Power and Energy*, Vol.95, No.2, pp.73–80 (1975) (in Japanese)
- (19) H. Obara, H. Oguri, and Y. Sato: “Theoretical Analysis of Harmonics in PWM Output Waveform of Multi-Level Inverters”, *IEEE Trans. IA*, Vol.134, No.1, pp.59–67 (2013) (in Japanese)

Shuhei Shimmyo (Member) received his B.E. degree in system design engineering and his M.E. degree in integrated design engineering from Keio University, Yokohama, Japan, in 2009 and 2011, respectively. He is currently with the Power and Industrial Systems Research and Development Center, Toshiba, Tokyo, Japan. His research interests include motion control and power electronics.



Kazuaki Yuki (Member) received his B.E. degree in electrical engineering and his M.E. degree in electrical engineering from Keio University, Yokohama, Japan, in 1992 and 1994, respectively. He is currently with the Power and Industrial Systems Research and Development Center, Toshiba, Tokyo, Japan. His research interests include motor drives, power electronics and railway systems.



Kouhei Ohnishi (Fellow) received his B.E., M.E., and Ph.D. degrees in electrical engineering from the University of Tokyo, Tokyo, Japan, in 1975, 1977, and 1980, respectively. Since 1980, he has been with Keio University, Yokohama, Japan. His research interests include mechatronics, motion control, robotics, and haptics. Prof. Ohnishi received Best Paper Awards from the Institute of Electrical Engineers of Japan and the Japan Society for Precision Engineering and Outstanding Paper Awards at IECON’85, IECON’92, IECON’93, and IECON’05. He also received the EPE-PEMC Council Award and the Dr.-Ing. Eugene Mittelman Achievement Award from the IEEE Industrial Electronics Society in 2004.

